

Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/ MICROBIOLOGY)

SEMESTER II EXAMINATION

BT707/MI707: BIOPROCESS TECHNOLOGY

Date: 30.04.2010

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in Part A should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of Part B in separate answer sheets.

PART A

Total marks: 20

Q1. Choose the correct option and put $\checkmark$ mark in front of it:

1. Agar-agar is obtained from
(a) Algae (b) bacteria (c) fungi (d) all of the above
2. For production of griseofulvin and chlortetracycline the presence of following salt is necessary in the fermentation medium
(a) KCl (b) KBr (c) KI (d) none of the above
3. The thermal destruction of essential media components follows
(a) second order kinetics (b) first order kinetics
(c) zero order kinetics (d) none of the above
4. In a fermentation process if the foaming observed is initially low and subsequently rises with time, then the foaming observed could be due to
(a) medium only (b) microbial activity only
(c) aeration (d) both medium and microbial activity
5. One of the aspect ratio given below CANNOT be the aspect ratio for an STR.
(a) 1:1 (b) 2:1 (c) 3:1 (d) 10:1
6. The following impeller can be used for sludge mixing
(a) horse shoe (b) rushton turbine (c) marine propeller (d) Scaba
7. Specific oxygen uptake rate increases with increase in the dissolved oxygen concentration up to a certain point known as
(a) t_c (b) C_{crit} (c) T_m (d) K_{La}
8. During mixing in a fermentor the value of V_{pump} depends on
(a) stirrer diameter (b) height of the fermentor
(c) diameter of the fermentor (d) t_m

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9. For microbial growth in fermentation process $\mu \approx \mu_{\max}$ is true for the following phase of the growth
- (a) lag (b) growth (c) stationary (d) death
10. For a fluid if $n > 1$ then the fluid is known as
- (a) pseudoplastic (b) bingham plastic (c) dilatant (d) cassan body
11. Fermentation broth of *Penicillium chrysogenum* **DOES NOT** exhibit following type of rheological behaviour
- (a) pseudoplastic (b) bingham plastic (c) cassan body (d) dilatant
12. If the K_s values of *Saccharomyces sp.*, *Escherichia coli* and *Aspergillus sp.* is 25 mg l^{-1} , 4.0 mg l^{-1} and 5.0 mg l^{-1} respectively, then the affinity of the organisms for glucose in a fermentation medium will be in the order of
- (a) *Saccharomyces* > *Escherichia* > *Aspergillus* (b) *Escherichia* > *Saccharomyces* > *Aspergillus*
(c) *Escherichia* > *Aspergillus* > *Saccharomyces* (d) *Aspergillus* > *Saccharomyces* > *Escherichia*
13. The expression $\mu = D$ applies for the following mode of cultivation of microorganisms
- (a) batch culture (b) continuous culture (c) fed batch culture (d) none of the above
14. The feed strategy in a fed batch fermentation system is decided on the basis of
- (a) q_p and μ (b) q_p and D (c) q_p and S (d) none of the above
15. Using the equation $\ln x_t = \ln x_0 + \mu t$, the specific growth rate of a bacterial culture can be calculated from the slope of a plot of
- (a) $\ln x_t$ against t (b) $\ln x_t$ against $\ln x_0$ (c) $\ln x_t$ against μ (d) x_t against x_0
16. In a fixed volume fed-batch culture, as biomass increases the specific growth rate will
- (a) increase (b) remain constant (c) decrease (d) none of the above
17. The following operation is **NOT** a component of downstream processing:
- (a) concentration (b) purification (c) cultivation (d) formulation
18. The required purity of the final product depends on its:
- (a) application (b) cost (c) quantity required (d) none of the above
19. The following procedure for disruption of microbial cells involves the use of liquid shear due to sudden release of pressure
- (a) ultrasonication (b) osmotic shock (c) enzyme lysis (d) microfluidization
20. The following is an example of liquid-liquid extraction procedure that requires the product to react with extraction liquid:
- (a) physical extraction (b) dissociative extraction (c) reactive extraction (d) all of the above

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SEMESTER II EXAMINATION

~~BT708/MT708~~ INDUSTRIAL BIOTECHNOLOGY

Instructions to the candidates:

1. Attempt all questions
2. Write answers for Section I and Section II of Part B in separate answer sheets.

707 - Bioprocess technology

Part B

Total Marks: 50

Section I

Q.1(A) Discuss the use of animal cells in bioprocessing. (02)

OR

Q.1(A) Enlist the properties required for a microorganism to be used in fermentation processes (02)

Q.1(B) Give criteria for formulation of industrial scale media. Discuss the use of whey as a fermentation substrate. (04)

OR

Q.1(B) Giving examples discuss the use of precursors and inducers in fermentation media. (04)

Q.1(C) Write a short note on **any two** of the following: (04)

- (i) antifoams
- (ii) sulfite waste liquor
- (iii) preservation strategies for industrial cultures
- (iv) role of phosphate in fermentation media

Q.2(A) Answer **any three** of the following: (03)

- (i) Classify fermentors on the basis of energy input and flow.
- (ii) Give schematic diagram for "gasket seal" in a fermentor.
- (iii) Enlist cardinal design rules for design of fermentors.
- (iv) Draw a fermentor with one multiblade impellor and label its parts.
- (v) Give schematic diagrams for major types of flow patterns generally observed in a stirred tank reactor

Q.2(B) What is ∇ factor? Discuss how "holding time" can be calculated for a batch sterilization process? (05)

OR

Q.2(B) What is $K_L a$? Describe the *sulfite oxidation technique* for determination of $K_L a$. (05)

Q.2(C) Answer **any one** of the following questions: (02)

- (i) Write a short note on: mixing time
- (ii) Explain: (i) V_{pump} (ii) circulation time (t_c)

OR

Q.2(C) Answer **any one** of the following questions: (03)

- (i) Giving labeled diagram discuss "sampling port" in a stirred tank fermentor.
- (ii) Write a short note on: Sparger
- (iii) Describe with schematic diagram: Cooling systems in a fermentor

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Section II

Q.3(A) What are non-Newtonian fluids? Giving rheogram and suitable example discuss pseudoplastic rheology. (05)

OR

Q.3(A) What is balanced growth? Giving Monod equation for balanced growth discuss the kinetics of product formation. (05)

Q.3 (B) Explain **any five** of the following terms in one or two sentences: (05)
(i) Y_{xs} (ii) Y_{ps} (iii) Y_{ATP} (iv) Y_{kcal} (v) RQ (vi) flow index
(vii) gross yield (viii) yield stress

Q.4(A) Answer **any two** of the following questions: (04)
(i) How is a quasi steady state attained in a Fed batch culture?
(ii) Explain the co-relationship between D_{crit} and wash-off of a microbial culture during continuous culture.
(iii) Write a note on applications of Fed batch culture.

Q.4 (B) Differentiate between **any two** of the following: (04)
(i) Batch and continuous culture
(ii) Steady state and quasi steady state
(iii) Variable Volume and fixed volume fed batch culture

Q.4(C) Justify **any one** of the following: (02)
(i) Despite high productivity and ease of operation in a continuous process, batch process is preferred for large scale production over continuous process.
(ii) An organism with a low K_s value for a substrate will have a long deceleration phase.

Q.5(A) Write a short note on **any two** of the following: (06)
(i) Conditioning of broth for solid-liquid separation
(ii) Non-mechanical disruption of whole cells
(iii) Adsorption to chromatographic particles

Q.5(B) Differentiate between **any two** of the following: (04)
(i) Evaporation and pervaporation
(ii) Extraction and perstraction
(iii) Homobifunctional and heterobifunctional ligands